

Stakeholder

13+3 · Buy / risk / decision

Completes JEPA arc: idea → scale/robot → tiny stable AC WM. Hook: 0.98 s planning. Teachable Alg.3. Required companion after I-JEPA + 2-AC for stability/latency.

SPINE: Stable E2E JEPA WM · MSE+SIGReg only · ~15M · 48× vs DINO-WM (0.98s/47s) · SIBLING to Meta JEPA · SIM-ONLY · PushT 96±2.83 · fixed-FLOP 90vs13 · recon hurts · λ -only

Talk arc — Stakeholder

Buy → hook → risks → decision

02-04

Why buy + taxonomy

JEPA arc · Fig.2 three-way · Alg.3 teachable

05-06

0.98s hook + control

Fig.3 speed · Fig.6 SR honesty

07-09

Risks + decision

sim-only · TwoRoom/Cube · H=5 · companion ask

10-13

Placement + tattoo + close

after I-JEPA+2-AC · exam bite

Why buy — schedule for Accompanying Models

Completes the JEPA lecture arc

Completes the arc

I-JEPA (idea) → V-JEPA 2.1/2-AC (scale + real robot) → LeWM (tiny AC WM without brittle heuristics).

Student pain killer

Answers “JEPA collapses / needs EMA / six loss knobs” with λ -only + SIGReg.

Systems hook

0.98 s vs 47 s · 15M on one GPU — syllabus-friendly vs Meta-scale pretrain.

Three-way map

Fig.2: E2E JEPA-WM vs foundation-freeze vs reward/recon WMs — teachable in one slide.

Open + Alg.3

lucas-maes/le-wm + website; Alg.3 fits one blackboard / one slide.

Fig. 2 — Taxonomy of latent WMs

E2E (PLDM→LeWM) · foundation freeze (DINO-WM) · generative/reward foils



Figure 2: **Characteristics of latent world model approaches.** Methods are grouped by training paradigm. *End-to-end* methods (PLDM) learn the encoder and predictor jointly from pixels without pre-trained representations or heuristics like stop-gradient or EMAs, but require many hyperparameters and lack collapse guarantees. *Foundation-based* methods (DINO-WM) avoid collapse by freezing a pre-trained vision encoder, forgoing end-to-end learning. *Task-specific* methods (Dreamer, TD-MPC) require reward signals or privileged state access. LeWM combines the strengths of all three: end-to-end, task-agnostic, pixel-based, reconstruction- and reward-free, with a single hyperparameter with provable anti-collapse guarantees.

1 Introduction

A central goal of artificial intelligence is to develop agents that acquire skills across diverse tasks and environments using a single, unified learning paradigm—one that operates directly from sensory inputs of its surroundings—without hand-engineered state representations or domain-specific calibration. Vision is ideally suited for this aim: cameras are inexpensive and scalable, and learning from pixels enables fully end-to-end training from raw sensory input to action [1]. World Models (WMs) are a powerful family of methods [2] that learn to predict the consequences of actions in the environment. When successful, WMs allows agents to plan and to improve themselves solely from their model of the world, i.e., in imagination space. This is particularly valuable in the offline setting, where agents must learn from fixed datasets without environment interaction—leveraging the model to generate synthetic experience and evaluate counterfactual action sequences [3, 4].

A recent popular approach for learning world models is the Joint Embedding Predictive Architecture (JEPA) [5]. Instead of attempting to model every aspect of the environment, JEPA focuses on

Positioning product for syllabus: where LeWM sits among WM families.

Hook — 0.98 s planning

48x vs DINO-WM · fixed-FLOP 90 vs 13 Push-T · teachable compute story



Figure 3: **Planning time and performance under fixed compute.** **Left:** Planning time comparison averaged over 50 runs. Encoding observations with $\sim 200\times$ fewer tokens than DINO-WM allows LeWM to achieve planning speeds comparable to PLDM while being up to $\sim 50\times$ faster than DINO-WM. **Center-Right:** Planning performance under the same computational budget (fixed FLOPs). LeWM significantly outperforms DINO-WM on Push-T (center) and OGBench-Cube (right). See App. D for planning setup details.

- We evaluate physical understanding in the latent space through probing of physical quantities and a violation-of-expectation test for detecting unphysical trajectories.

2 Related Work

World Models aim to learn predictive models of environment dynamics from data, enabling agents to reason about future states in imagination. A prominent class of WMs consists of *generative* approaches that explicitly model environment dynamics in pixel space. These action-conditioned generative models act as learned simulators by producing future observations conditioned on past

Lead with systems number; then qualify sim-only and TwoRoom honesty.

Teachable recipe — Alg. 3 in one slide

What students can re-implement overnight

1 Encode $o_t \rightarrow z_t$ (ViT-Tiny + BN projector)

2 Predict $\hat{z}_{t+1} = \text{pred}(z_t, a_t)$ via AdaLN transformer

3 Loss = MSE + λ SIGReg — only λ to tune

4 No EMA · no stop-grad · no freeze · no recon

5 Plan: CEM on $\|\hat{z}_H - z_g\|$ · $H=5$ · MPC replan

6 ~15M params · single GPU · few hours

Risks / costs / limits

Honesty required for the buyer

Sim-only

No Franka / BEHAVIOR / DROID numbers. Do not sell as replacing V-JEPA 2-AC robotics lecture.

TwoRoom + Cube

Loses on simplest env; slightly trails DINO on rich 3D Cube — say it aloud.

Short horizon

H=5 / 25 steps; paper admits hierarchical WM needed for long horizon.

Action labels

Needs action-labeled offline coverage; low diversity hurts SIGReg.

Empiricist weight 3

Expect deep number grilling — Stakeholder empty seat still owns the slot pick.

“First” claim

Definitional boundaries vs concurrent LeJEPA / Causal-JEPA — don't oversell novelty alone.

Control honesty — Fig. 6

Push-T win · Reacher competitive · Cube trails DINO · TwoRoom behind

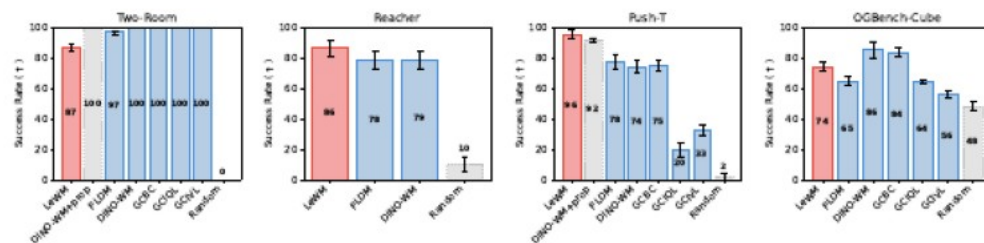


Figure 6: **Planning performance across environments.** Results are shown for *Two-Room* (left), *Reacher* (center 1), *PushT* (center-2) and *OGBench-Cube* (right). LeWM consistently outperforms PLDM and DINO-WM on Push-T and Reacher. On OGBench-Cube, DINO-WM slightly outperforms LeWM, possibly due to the higher visual complexity and the 3D nature of the environment, which makes encoder training more challenging. In the simpler Two-Room environment, PLDM and DINO-WM outperform LeWM, which may be explained by the SIGReg regularization encouraging a Gaussian distribution in a high-dimensional latent space, while the intrinsic dimensionality of the environment is much lower.

DINO-WM. Notably, on PushT, LeWM (pixels-only) surpasses DINO-WM even when DINO-WM has access to additional proprioceptive information, demonstrating LeWM’s ability to capture underlying task-relevant quantities. Interestingly, LeWM performs worse on the simplest environment, Two-Room. A possible explanation is that the low diversity and low intrinsic dimensionality of this dataset make it difficult for the encoder to match the isotropic Gaussian prior enforced by SIGReg in a high-dimensional latent space, which may lead to a less structured latent representation. This highlights a potential limitation of the SIGReg regularization in very low-complexity environments.

Moreover, when comparing planning speedups (Fig. 3), LeWM achieves a $48\times$ faster planning time, with the full planning completing in under one second while preserving competitive performance across tasks. This planning time remains consistent across environments for a fixed planning setup, narrowing the gap toward real-time control.

4.3 Towards Stable Training of World Models

Ablations. We perform ablations on several design choices of LeWM. First, we analyze the sensitivity of SIGReg to its internal parameters, namely the number of random projections and the number of integration knots. The performance is largely unaffected by these quantities, indicating that they do not require careful tuning. As a result, the regularization weight λ remains the only effective hyperparameter. Since only a single hyperparameter needs to be tuned, grid search can be performed efficiently using a simple bisection strategy ($\mathcal{O}(\log n)$), whereas PLDM requires search in polynomial time ($\mathcal{O}(n^6)$). We also study the effect of the embedding dimensionality. While the representation dimension must be sufficiently large for the method to perform well, performance

Lead with Push-T 96 ± 2.83 ; immediately disclose TwoRoom/Cube.

Decision ask

What to schedule after this paper club

REQUIRED COMPANION

After I-JEPA + V-JEPA 2-AC: schedule LeWM for the “training stability & planning latency” module.
Pair with PLDM / DINO-WM as foils.

OPTIONAL

Optional if the track only covers Meta SSL → Franka. Still cite LeWM when students ask how to train a tiny AC WM without EMA / six knobs.

Decision: required companion for stability/latency · not a 2-AC replacement

Placement in the syllabus

SIBLING module · not Meta descendant lecture

I-JEPA

Birth certificate — predict embeddings

CLEAR prior

V-JEPA 2.1 / 2-AC

Scale + dense unlock + Franka CEM

CLEAR robot path

LeWM (this)

Tiny stable E2E AC WM · λ -only · 0.98s

SIBLING recipe

PLDM / DINO-WM

Foils: fragile multi-term · frozen foundation

CLEAR foils

Exam bite — Stakeholder

Say this in 30 seconds

LeWM is the teachable companion that closes the JEPA arc: after the idea (I-JEPA) and the robot path (2-AC), students need a tiny, stable, end-to-end action-conditioned world model with one loss knob and sub-second CEM. Buy the slot for stability/latency — not as a Franka replacement. Disclose sim-only, TwoRoom, and short H=5.

Decision tattoo: REQUIRED companion after I-JEPA + 2-AC · OPTIONAL if Meta-only track.

Numbers tattoo — pin these

Shared spine across all five seats

~15M

params · 1 GPU / few h

0.98s vs 47s

~48× planning speed

~200×

fewer tokens vs DINO

90 vs 13

fixed-FLOP Push-T

74 vs 48

fixed-FLOP Cube

96.0±2.83

Push-T SR (Tab.5)

78±5 / 92±1.63

PLDM / DINO Push-T

$\lambda \in [0.01, 0.2]$

SR>80%; peak ~0.09

96 → 86

recon loss hurts (T7)

SIBLING

not Meta descendant

SIM-ONLY

cite 2-AC for robots

λ -only

vs PLDM 6 knobs

Stakeholder — close

Takeaways + exam bite

1. Buy: completes JEPA arc; 0.98s hook; Alg.3 teachable; open code.
2. Risks: sim-only; TwoRoom/Cube honesty; short H=5; action-labeled data.
3. Decision: required companion for stability/latency after I-JEPA + 2-AC.
4. Do not replace V-JEPA 2-AC robotics lecture with LeWM.

EXAM BITE

Schedule LeWM as the tiny stable AC WM module; cite 2-AC for real robots.